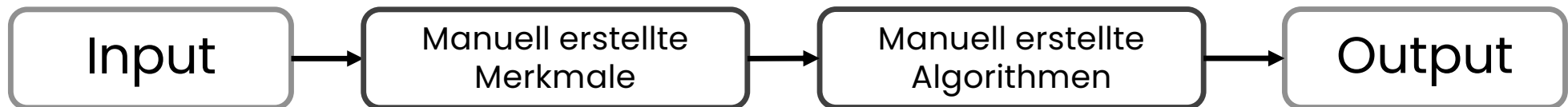


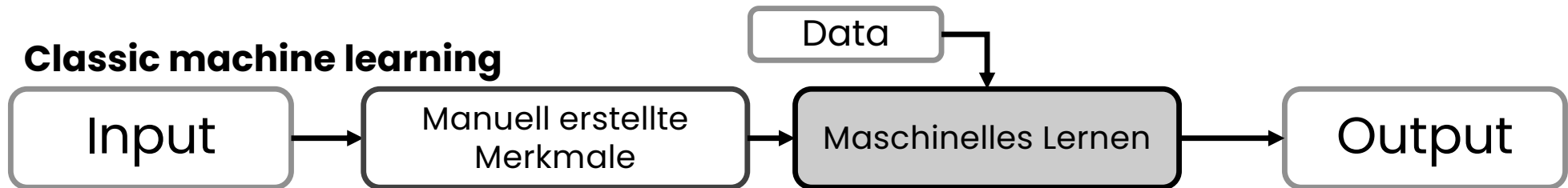
Deep Learning models

Different approaches

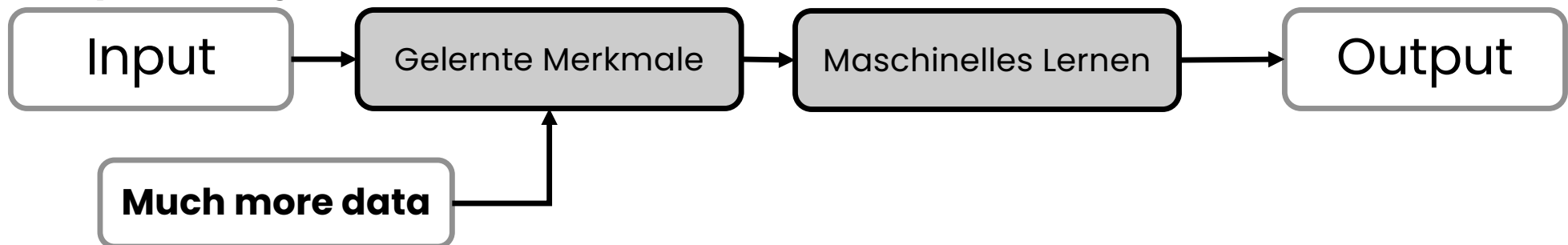
Classic image processing



Classic machine learning



Deep Learning



Inspired by Figure 5.2 in book by Szeliski, 2021

Image classification

Input: an image

Output: the class name for this image

The class label is generally one or more of the discrete labels used during training, i.e.

{cat, dog, cow, toaster, apple, tomato, truck, ... }

```
def classifier(image):  
    //Do some stuff  
    return class_label;
```

$f(\text{) = \text{"Cat"}$

$f(\text{) = \text{"Dog"}$

$f(\text{) = \text{"Toaster"}$

Semantic gap



What humans see

```

)1110 111010010111
1010111 110101010(
1 01001 1111010111(
10101011110111011(
1001111001 1 01 1(
1010111 1111101101(
101 11101 100100101
10011111 1000111001
1010010010011100011
10000 100011101110(
01100000001111 1 1(
1 1011001 010011 1(
11 10001 1111 1011
111 101010 10111 1
1000101010010101101
1001 1111000010 11(
10111100001111 1101
    
```

What computers see

```

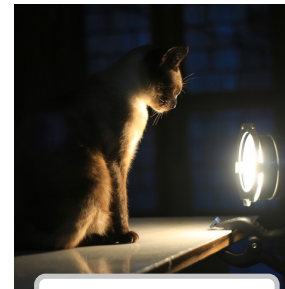
,1 1101 11010
,0111 0110011
111110010110
,011000010 10
100 001101111
1100 0 011 0
,10111000 110
100010 111 1
,01010111101
1 11101011 01
, 11111011001
,1 11100111 1
,1110010 1011
,000001110100
101001 001000
,010110110011
,01110011 110
,1111 1111011
,110111001111
,111111110 00
1010110100 1
10 0100111101
1011 10101111
    
```

Variation makes classification hard

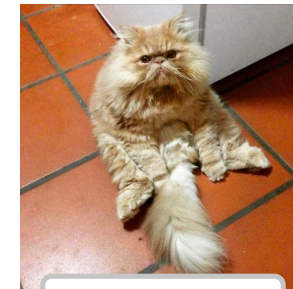
- The same class of objects can appear very different in different images.



Perspective



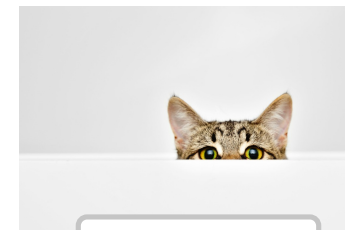
Illumination



Deformation



Background clutter



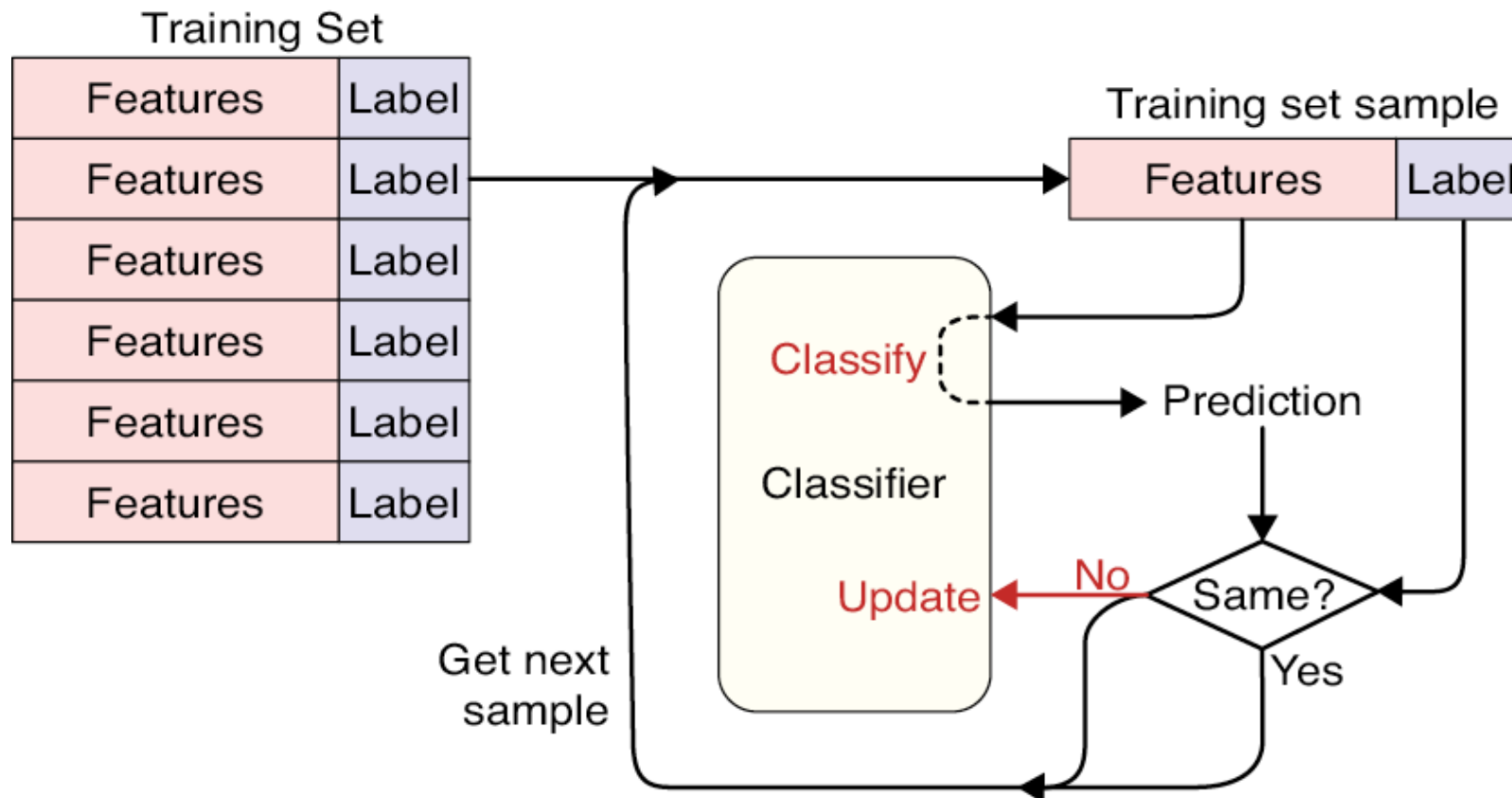
Occlusion

The problem is ill-defined

- Different realities can produce the same picture ...
- It is generally impossible to calculate the “correct” answer, only the most probable one ...
- You need prior knowledge on which to build. This can be learned from data.

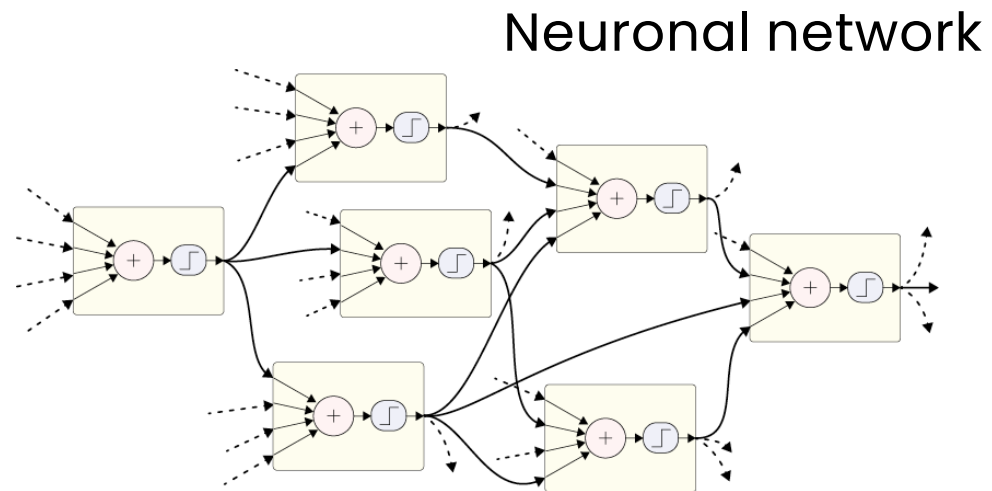
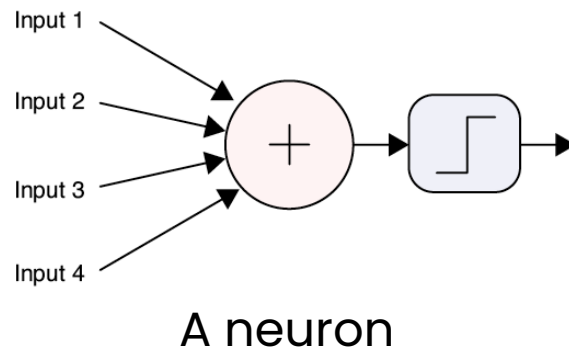


Training process



From Andrew Glassner, Deep Learning - a visual approach

Neuronal networks



From Andrew Glassner, Deep Learning – a visual approach